

Ahmed Hanye Hossny

Full-Stack Software Engineer

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Summary

Full-Stack Software Engineer and Computer Science graduate specializing in resilient local-first architectures, sub-millisecond client reactivity, and scalable backend systems across TypeScript and Python. Proven track record architecting offline-first synchronization engines, high-throughput asynchronous worker pipelines (Django Ninja + Celery + Redis), and modular Next.js / React web applications with strict type safety and automated testing gates.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Bash/Shell Scripting, HTML5 & CSS3

Frontend Engineering: React, Next.js (App Router), Tailwind CSS, TanStack Query, TanStack Form, Zod, Three.js / R3F, GSAP, shadcn/ui, @base-ui/react, next-intl

Backend & Distributed: Django (Ninja, REST), Celery (Multi-lane Workers), Redis, RESTful APIs, AWS SES, Headless Session Auth, CSRF Protection

Databases & Sync: PostgreSQL, SQLite (wa-sqlite WASM), MySQL, PowerSync (WAL Streaming Sync), TanStack DB (In-Memory Reactive Store)

DevOps & Architecture: Docker & Docker Compose, Kubernetes, Git & GitHub Actions (CI/CD), Linux CLI, Local-First Architecture, Feature-Sliced Design (FSD)

Testing & Quality DX: Playwright (E2E), Vitest, Pytest, Storybook, Bun, uv, Ruff, ESLint, React Doctor, Fallow, Steiger

Education

Arab Open University (AOU), B.Sc. in Computer Science

Sept 2022 – July 2026

- Core Coursework: Algorithms & Data Structures (M269), Software Engineering (TM354), Data Management & Analysis (TM351), Web Technologies (TT284), Operating Systems (TM298), Computer Organization & Architecture (TM103), Machine Learning & AI (TM358), Object-Oriented Programming (M251), Computer Graphics (TM240), Discrete Mathematics & Linear Algebra.

Technical Projects

InterviewFlow — Production Technical Hiring SaaS Engine

Nov 2025 – present

High-standard enterprise technical hiring platform engineered to conduct reliable, zero-latency live technical assessments and automated candidate scorecards (in active development toward MVP).

- Architected a dual-engine local-first client pairing TanStack DB for sub-millisecond ($< 1\text{ms}$) in-memory query reactivity with PowerSync and WASM SQLite (wa-sqlite) for background WAL streaming sync, achieving 100% evaluation data retention during network disconnects.
- Designed a high-throughput, decoupled REST API using Django Ninja with strict end-to-end OpenAPI type contracts and secure headless session cookie authentication with cross-origin CSRF defense.
- Built distributed Celery multi-lane asynchronous worker pipelines backed by Redis message brokers for transactional email delivery via AWS SES and background PDF assessment synthesis.
- Implemented Feature-Sliced Design (FSD) across the Next.js / React client with Steiger architectural linting, eliminating circular dependencies and ensuring strict layer isolation.
- Established automated quality assurance with Playwright E2E suites and Storybook interaction play tests, achieving 90%+ critical path coverage and zero regression deployments.

[InterviewFlow Sandbox — Graduation Project Prototype](#)

Sept 2025 – July 2026

Exploratory real-time candidate evaluation sandbox developed as an academic graduation project at Arab Open University, validating the feasibility for the v2 SaaS engine.

- Engineered a synchronized multi-evaluator scoring interface with live evaluation rubrics using React, TypeScript, Vite, Tailwind CSS, and shadcn/ui.
- Integrated Supabase and PostgreSQL with Row-Level Security (RLS) policies and optimistic client updates for concurrent interviewers.
- Implemented full test suites using Playwright and Vitest to validate real-time synchronization mechanics and scoring state persistence.

[Retro OS & 3D Interactive Portfolio Platform](#)

Jan 2026 – July 2026

Dual-experience portfolio combining an interactive 3D CRT workstation and a simulated Windows 95 desktop OS.

- Engineered a 3D workstation simulation with Three.js, React Three Fiber, and Spline, featuring dynamic cursor tracking and interactive lighting.
- Implemented a complete Windows 95 desktop environment in React / Next.js with draggable, resizable window managers and Web Audio API procedural soundscapes.
- Delivered full internationalization (English & Arabic) with RTL layout adaptation using next-intl, maintaining 60 FPS graphics and zero WebGL memory leaks.

[Awwwards Interactive Web Experience](#)

Mar 2025 – June 2025

High-framerate interactive web showcase demonstrating fluid scroll-driven storytelling and complex animation orchestration.

- Orchestrated 60 FPS interactive animations, parallax effects, and layout morphs using GSAP (@gsap/react) and ScrollTrigger.
- Built responsive, hardware-accelerated SVG polygon transitions and dynamic micro-interactions with Tailwind CSS.

Certifications

[GitHub Foundations](#)

GitHub / Credly

- Validated core Git workflows, repository collaboration, GitHub Actions CI/CD automation, and repository security best practices.

[AI Engineer for Developers Associate](#)

DataCamp

- Validated LLM APIs, LangChain workflows, OpenAI APIs, vector embeddings, and production AI integration.

[SQL Associate Certification](#)

DataCamp

- Verified relational schema modeling, PostgreSQL query optimization, multi-table joins, data transformation, and aggregation.

[Containerization & Virtualization with Docker & Kubernetes](#)

DataCamp

- Verified Docker container lifecycles, multi-stage builds, image optimization, Docker Compose, and Kubernetes orchestration.

[Python Developer Career Track](#)

DataCamp

- Validated OOP architecture, data structures & algorithms, package development, clean code principles, and pytest unit testing.

[The Full Stack](#)

Meta

- Verified full-stack web architecture connecting Django backend services with modern responsive frontend components and REST APIs.

[APIs](#)

Meta

- Validated RESTful API architecture, Django REST Framework serialization, token authentication, endpoint design, and API testing.

[Django Web Framework](#)

Meta

- Verified Django MVT architecture, ORM data modeling, migrations, user authentication, views routing, and admin interfaces.

[Back-End Developer Capstone](#)

Meta

- Implemented complete backend services with Django, Django REST Framework, MySQL database integration, and automated unit tests.

[Introduction to Databases for Back-End Development](#)

Meta

- Verified relational database modeling, SQL queries, MySQL database administration, table constraints, and schema design.

[Professional Front-End Web Development](#)

Udacity

- Accredited in advanced DOM manipulation, responsive web architecture, asynchronous Web APIs, and client-side performance.

[Learn Linux](#)

Boot.dev

- Validated Linux terminal proficiency, shell scripting, process management, file permissions, I/O redirection, and package management.

[Learn Git](#)

Boot.dev

- Validated Git internals, plumbing commands, branching workflows, interactive rebasing, merge strategies, and conflict resolution.